

2016.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TDV6 3.0L DIESEL

LEFT FUEL INJECTORS (G1509431)

REMOVAL AND INSTALLATION

19.60.09	INJECTORS	3000 CC, TDV6	2.4	USED WITHINS	+
	- LH BANK				
	SET - RENEW				

SPECIAL TOOL(S)



100-012
Slide Hammer



310-213

Fuel Injector
Removal Adapter

PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME
Installation Step 2	Renew Part	Fuel injector seals
Installation Step 2	Renew Part	Fuel injector washers
Installation Step 5	Renew Part	Fuel injector high pressure pipes

REMOVAL

WARNING:

Diesel fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is therefore essential that absolute cleanliness is observed when working with these components. Always install new blanking plugs to any open orifices or lines. Failure to follow this instruction may result in foreign matter ingress to the fuel injection system.

NOTE:

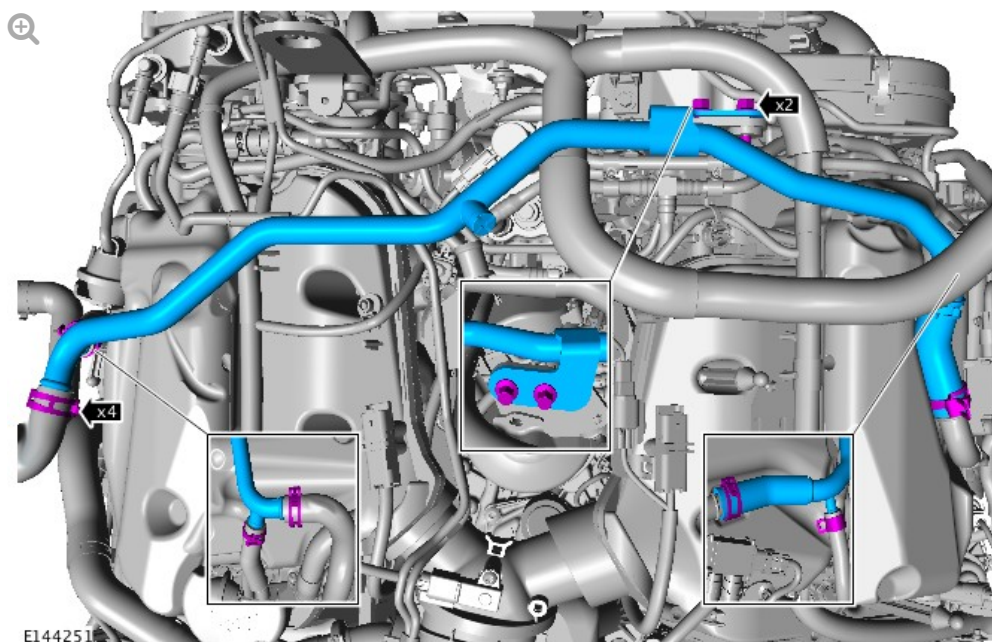
Some variation in the illustrations may occur, but the essential information is always correct.

1. Refer to: [Fuel Injection Component Cleaning](#) (303-04A Fuel Charging and Controls - TDV6 3.0L Diesel, General Procedures).
2. Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).
3. Refer to: [Secondary Bulkhead Centre Panel - TDV6 3.0L Diesel](#) (501-02 Front End Body Panels, Removal and Installation).
4. Refer to: [Cooling System Partial Draining and Vacuum Filling](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, General Procedures).

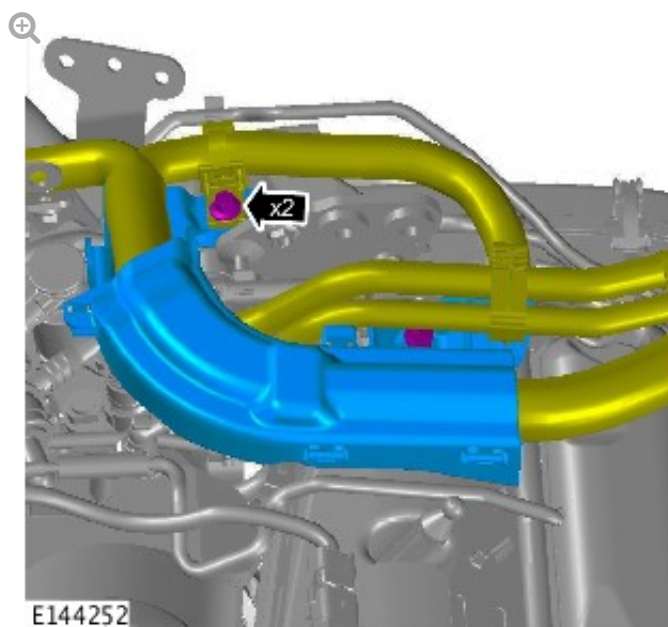
5.

CAUTION:

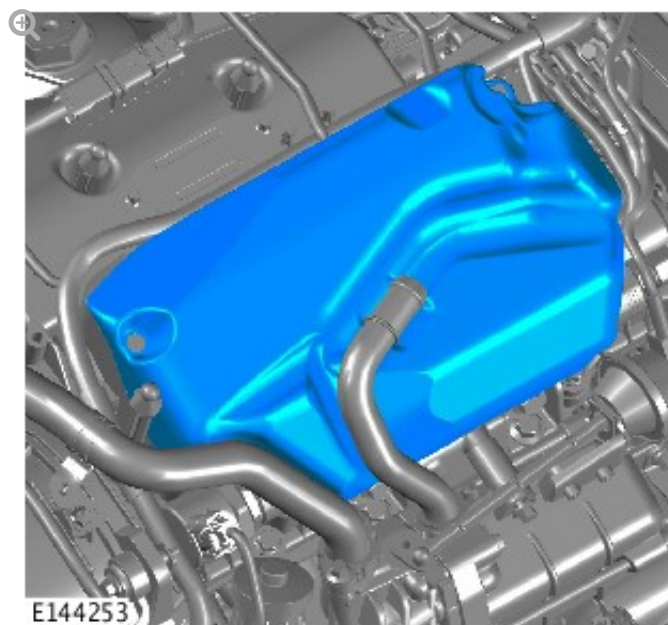
Be prepared to collect escaping coolant.



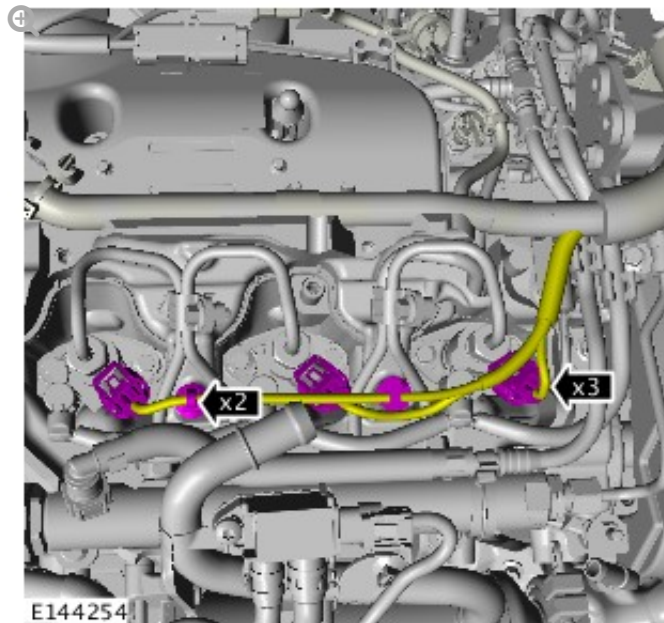
6.



7.



8.



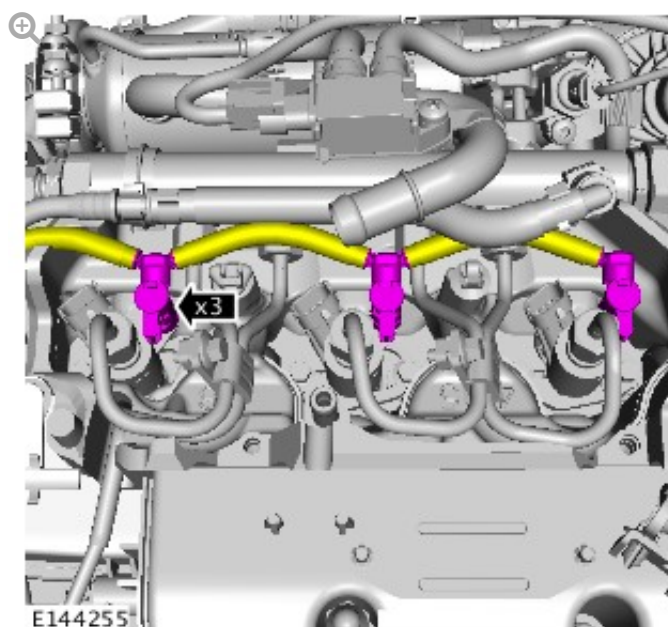
9.

CAUTIONS:

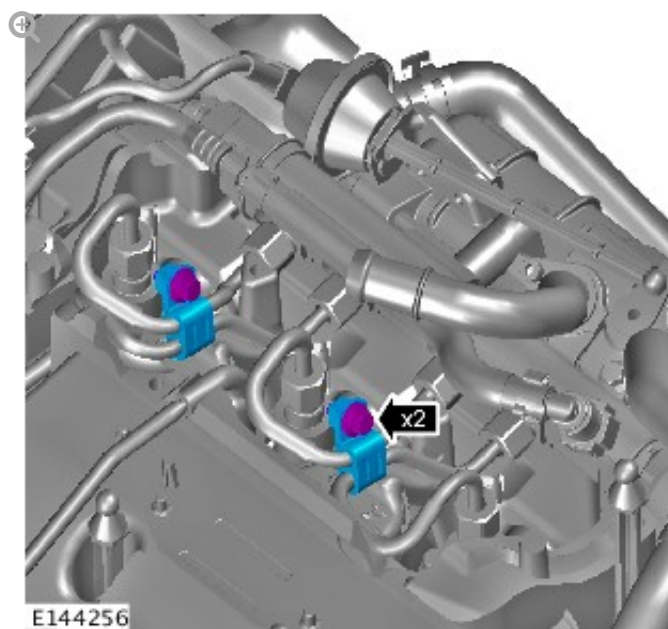
- Be prepared to collect escaping fuel.
- Before disconnecting or removing components, ensure the area around the joint faces and connections are clean. Plug open connections to prevent contamination.

NOTE:

Make sure that the fuel injector return line has a maximum of 8 uses.

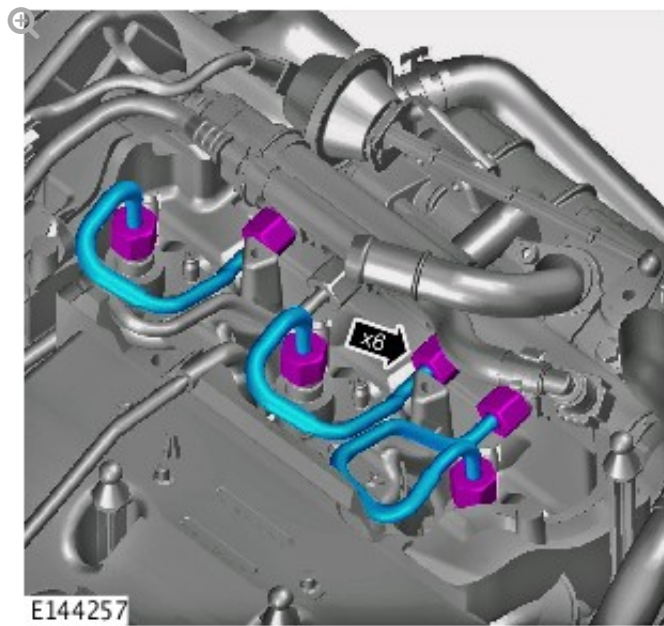


10.

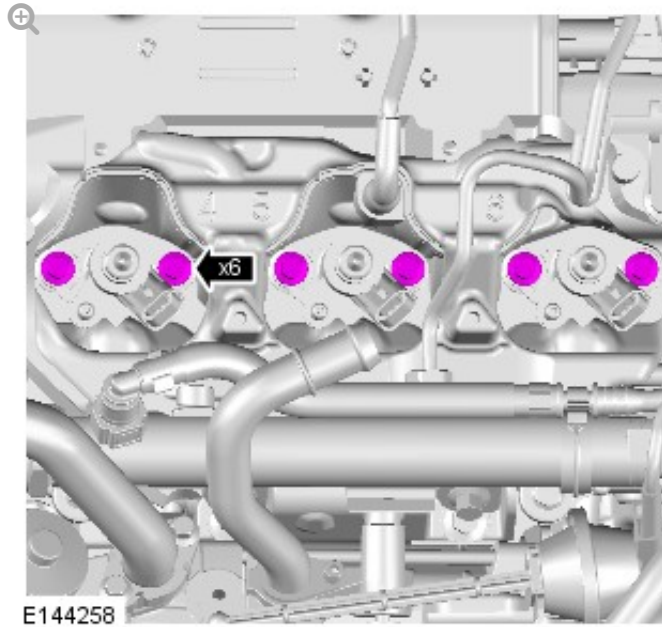


CAUTIONS:

- Be prepared to collect escaping fuel.
- Before disconnecting or removing the components, make sure the area around the joint faces and connections are clean.
- Remove and discard the high-pressure fuel supply lines.



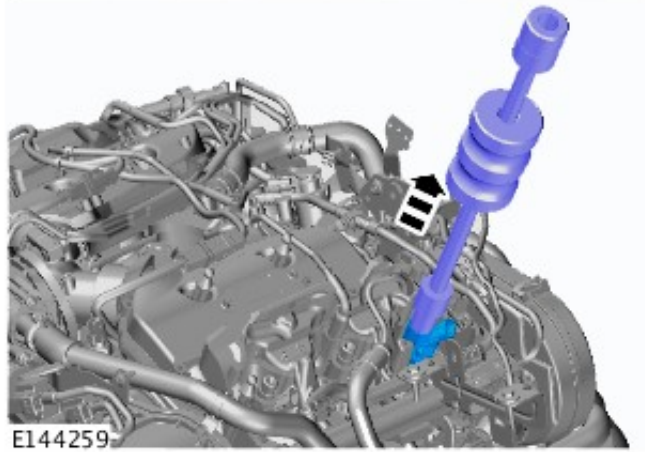
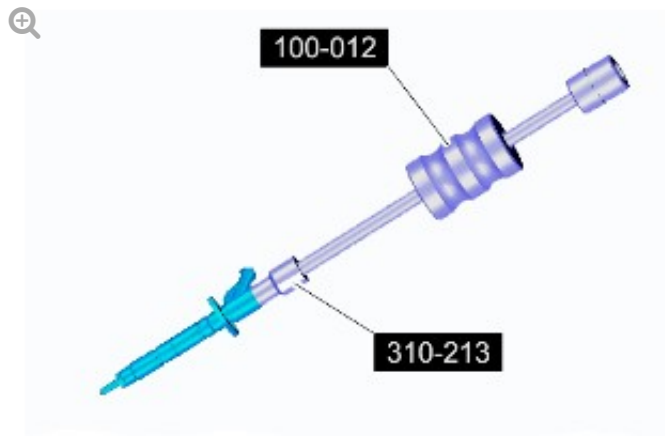
12.



13.

NOTE:

Repeat this step for each of the fuel injectors.

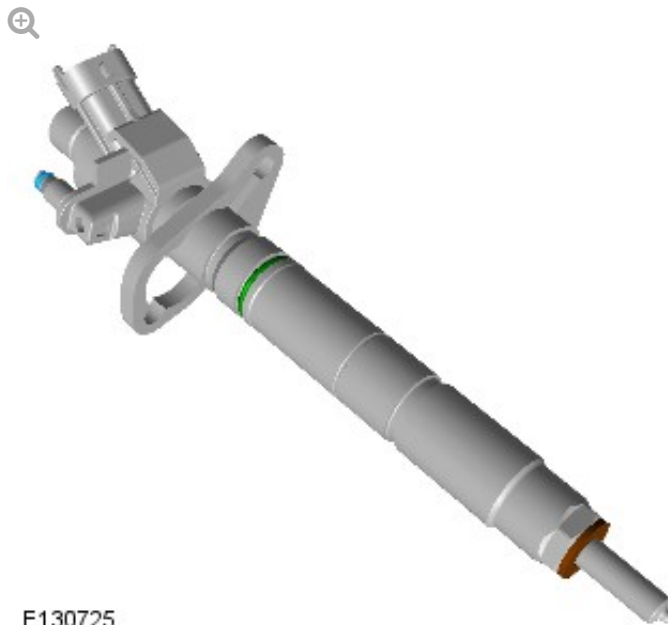


- *Special Tool(s):* 310-213
- *Special Tool(s):* 100-012

14.

NOTES:

- Remove and discard the O-ring seals.
- Discard the sealing washers.
- Repeat this step for each of the fuel injectors.

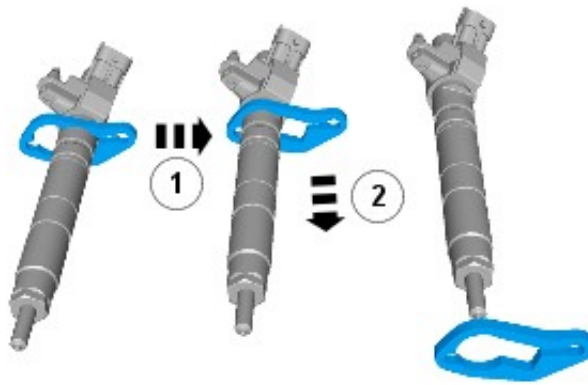


E130725

15.

NOTE:

This step is only required if new components are not installed.



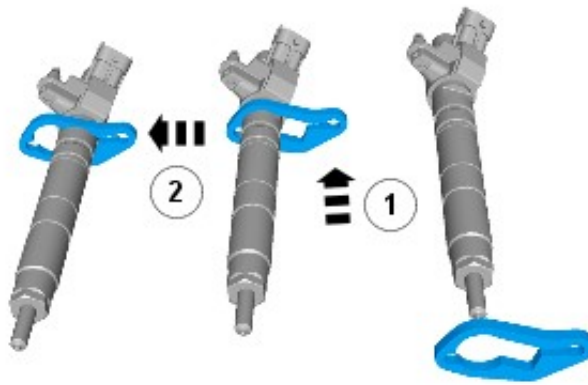
E144699

INSTALLATION

1.

NOTE:

This step is only required if previously removed.



E144700

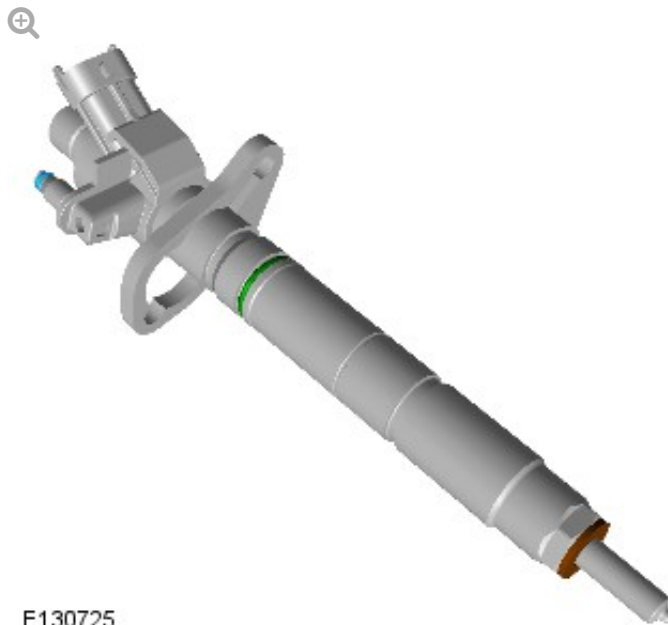
2.

CAUTION:

Make sure that new sealing washers are installed.

NOTES:

- This step is only required if previously removed.
- Install new O-ring seals.
- Repeat this step for each of the fuel injectors.



E130725

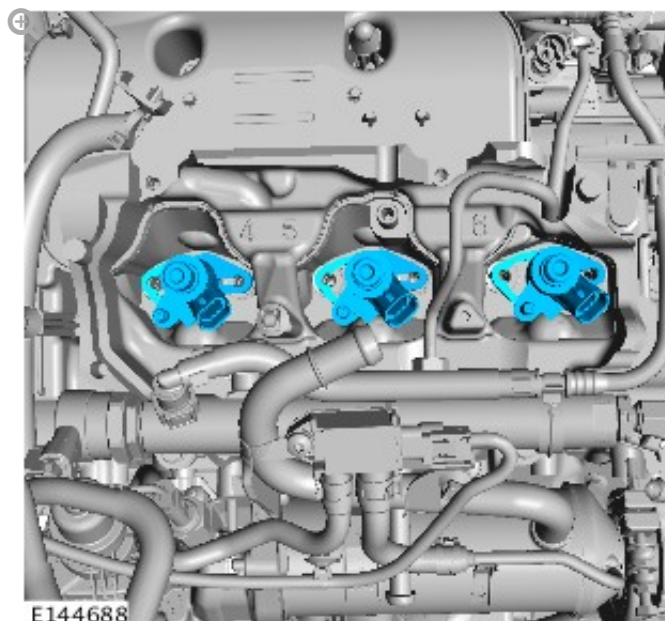
Renew Part: [Fuel injector seals](#).

Renew Part: [Fuel injector washers](#).

3.

CAUTION:

Make sure that the area around the open fuel injector ports are clean and free of foreign material and lubricant prior to installing the fuel injector.

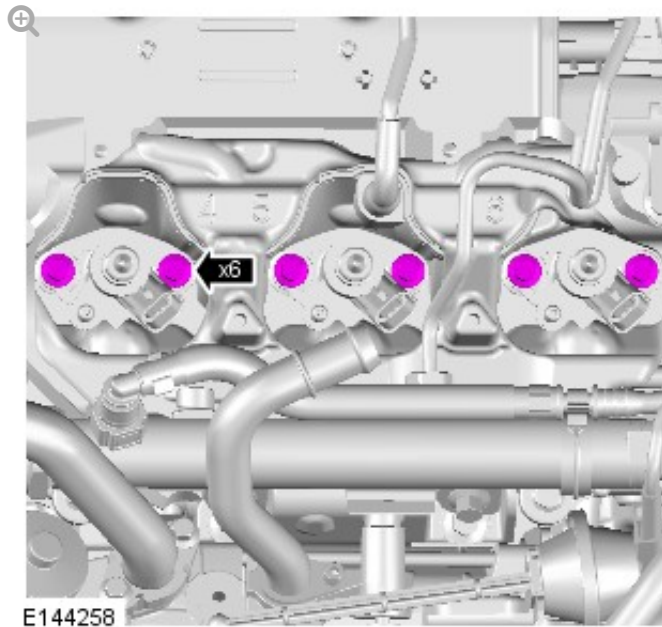


E144688

4.

NOTE:

Tighten the retaining bolts evenly and progressively.

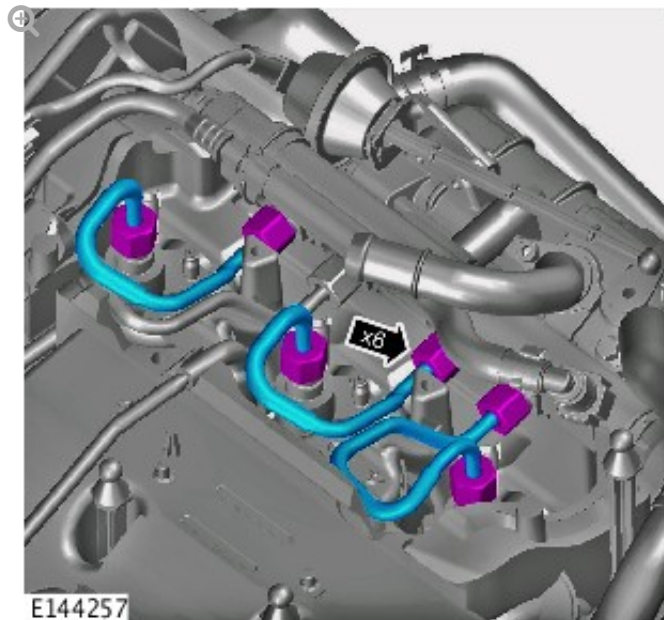


Torque: 9 Nm

5.

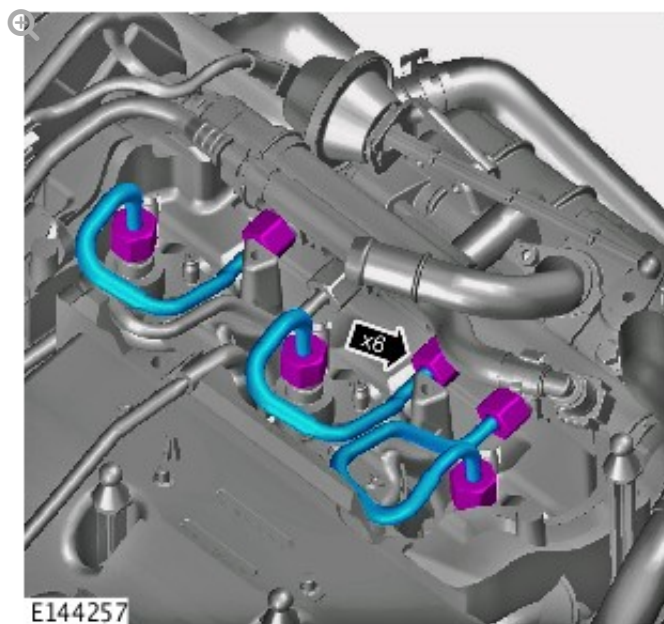
CAUTIONS:

- Make sure that a new component is installed.
- Only tighten the unions finger-tight at this stage.



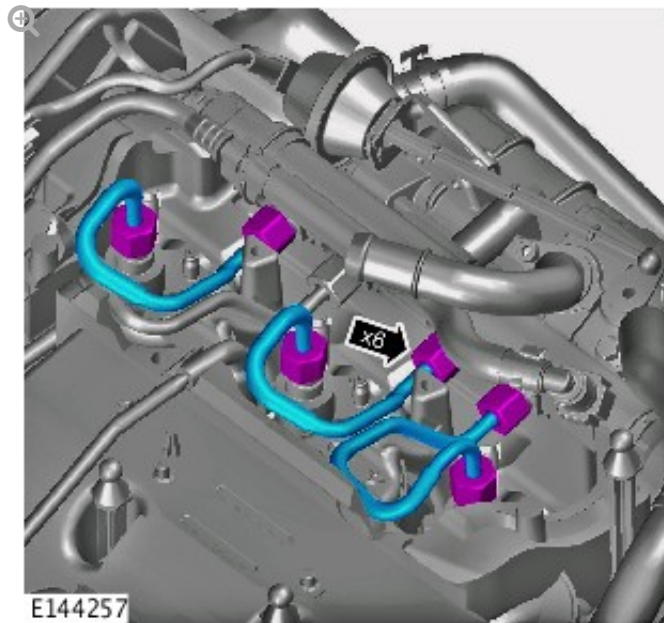
Renew Part: *Fuel injector high pressure pipes.*

6.



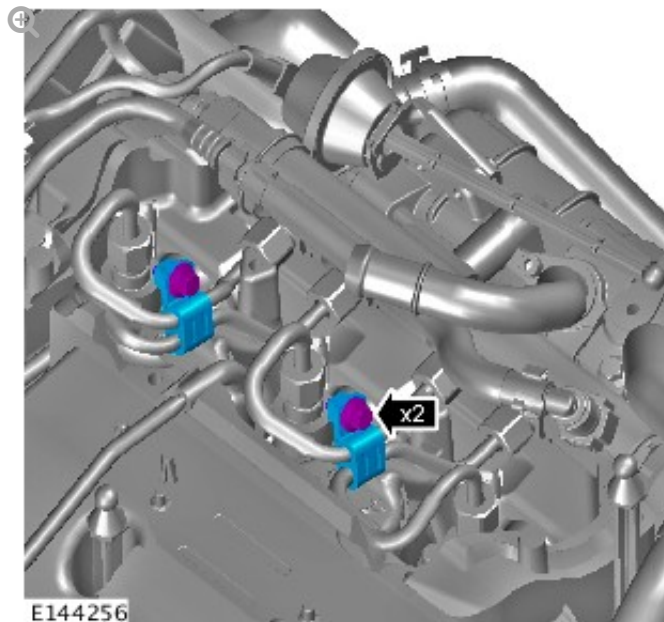
- Stage 1: Tighten the high-pressure fuel supply line unions at the fuel rail to 15Nm.
- Stage 2: Tighten the high-pressure fuel supply line unions at the injector to 15Nm.

7.



- Stage 1: Tighten the high-pressure fuel supply line unions at the fuel rail to 35Nm.
- Stage 2: Tighten the high-pressure fuel supply line unions at the injector to 35Nm.

8.

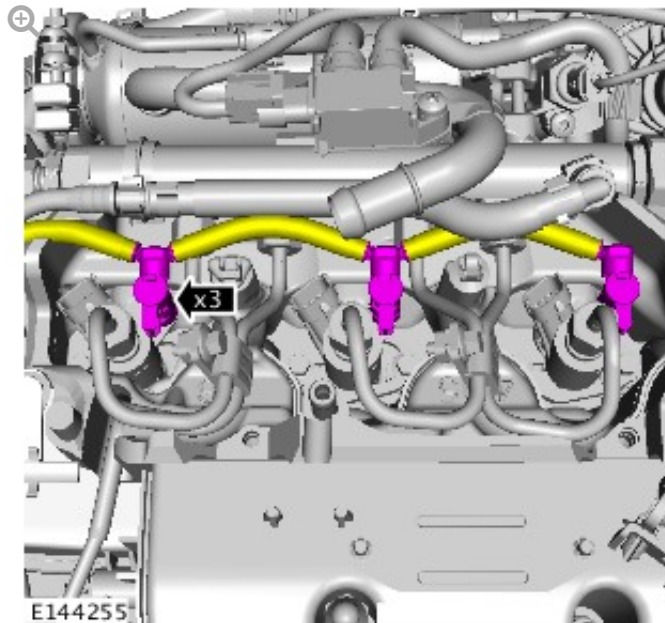


Torque: 10 Nm

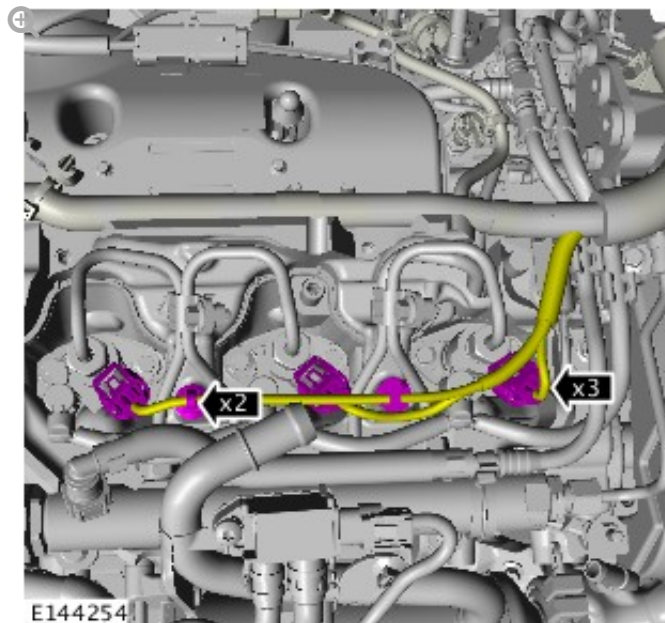
9.

NOTE:

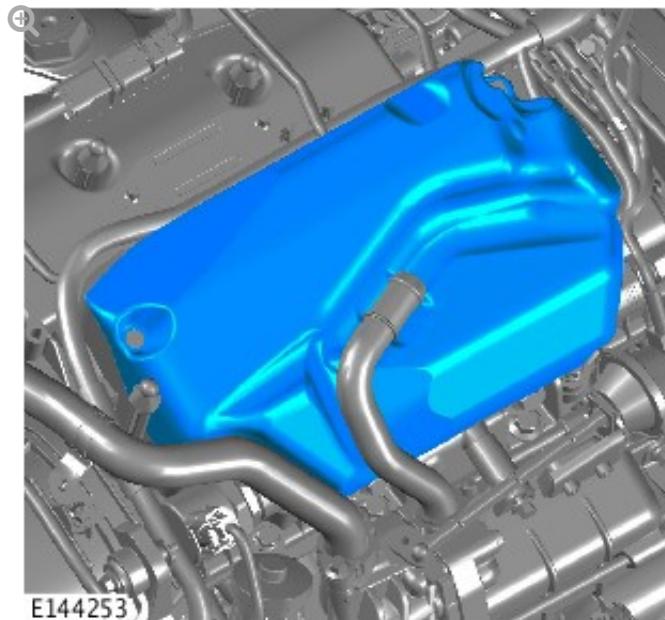
Make sure that the fuel injector return line has a maximum of 8 uses.



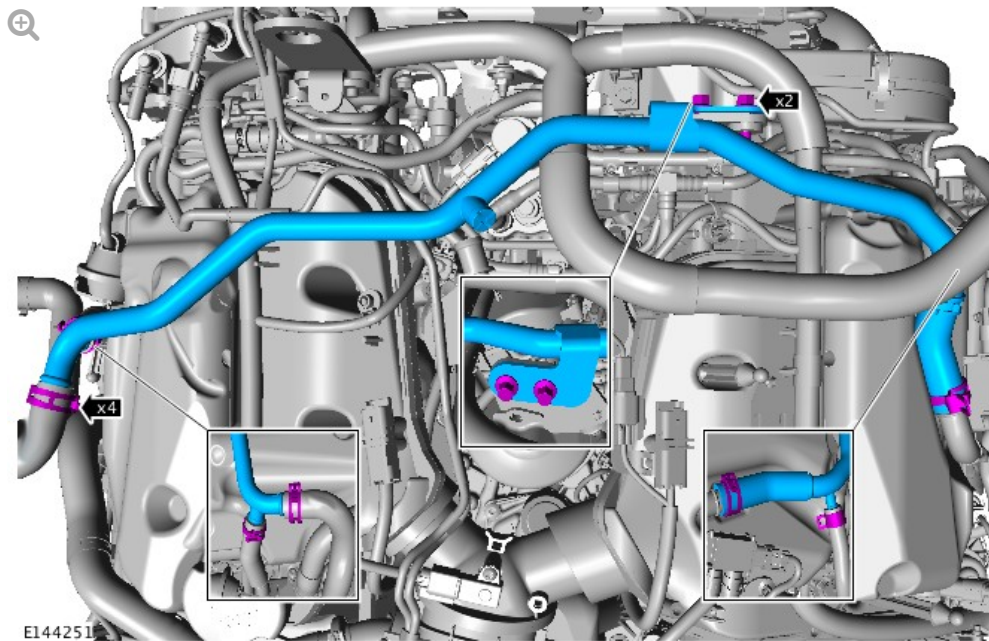
10.



11.



12.



Torque: 10 Nm

13. Refer to: [Cooling System Partial Draining and Vacuum Filling](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, General Procedures).
14. Refer to: [Secondary Bulkhead Centre Panel - TDV6 3.0L Diesel](#) (501-02 Front End Body Panels, Removal and Installation).
15. Refer to: [Specifications](#) (414-00 Battery and Charging System -

General Information, Specifications).

16. If a new unit is installed, configure using the approved diagnostic tool.